3D Imaging & Optical Sensing

SDlab
LAB OVERVIEW · 2026

SDlab is a research group at the School of Electronics Engineering, KIIT Deemed to be University. We build optical instruments that see in three dimensions, using digital holography, stereo imaging and fiber-optic sensors, and apply them to biology, water quality and the marine environment.

14 Journal publications	**4** Conference papers in 2026	**3** Indian patents (2 granted)	**1** SERB-funded project completed as PI	**Top 35** 5G Innovation Hackathon 2026 (2 teams)

RESEARCH AREAS

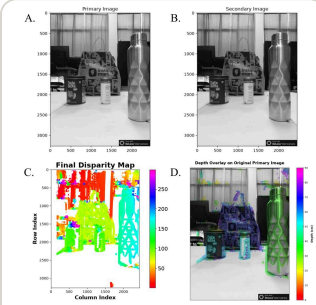
Digital holography
Digital holographic imaging for 3D visualization and characterization of biological and aquatic samples: in situ holography, 3D reconstruction and automated analysis of microscopic organisms.

Stereo imaging
Calibrated stereo systems for depth reconstruction, from in situ imaging of marine zooplankton to low-cost single-camera stereo using a smartphone.

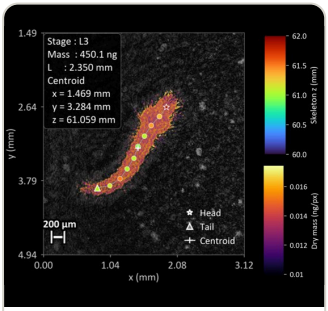
Optical fiber sensing
Fiber-optic and photonic crystal fiber sensors for acoustic, curvature, acceleration, humidity and refractive-index measurement, including plasmonic designs for bio-sensing.

AI-assisted imaging
CNN-based analysis and automated classification of microscopic organisms in large holographic and stereo image datasets.

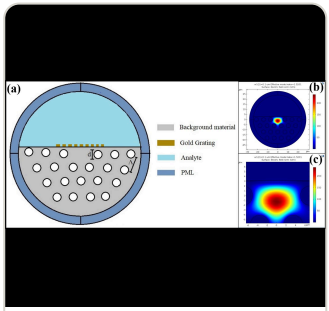
RECENT WORK



Single-camera 3D imaging. A smartphone on a 3D-printed sliding stand captures a stereo pair; a two-pass disparity algorithm recovers depth. Accepted, IEEE conference 2026.



Label-free holography of mosquito larvae. Inline digital holography of a live larva gives its 3D position, orientation, length and dry mass. Submitted, Photonics 2026, IIT Hyderabad.



Bio-compatible fiber sensor. D-shaped Zeonex photonic crystal fiber with a gold grating; surface plasmon resonance gives 10.95 nm/RIU sensitivity. Presented, JSAP–Optica 2026, Japan.



HoloBuoy (proposed). A solar-powered buoy carrying a submerged holographic imager for near real-time water quality monitoring, connected over 5G.

PRINCIPAL INVESTIGATOR



Dr. Sumit Dass

Principal Investigator, SDlab · School of Electronics Engineering, KIIT University

- Ph.D., IIT Bhubaneswar, in optical fiber sensing technology
- Postdoctoral research in optics at Harbor Branch Oceanographic Institute, Florida Atlantic University (USA), on holographic and stereo imaging of marine microorganisms
- Lead PI of a SERB (Govt. of India) project on IR absorption spectroscopy with a tapered PCF supercontinuum source
- Teaching: Optics, Optical Fiber Communication, Digital Logic Design, Satellite Communication, Network Analysis

STUDENTS



Debarchana Sen
4th yr ECS



Dipashrita Das
4th yr ECS



Saurav Jena
3rd yr EC



Saanvi Patankar
3rd yr EC



Narayan Aravind
3rd yr EC



Ustah Puspalak
3rd yr EC



Rushati Maji
3rd yr EC



Gurudutta Panda
2nd yr BioTech



Anushree Dash
2nd yr BioTech



Anirban Mandal
2nd yr BioTech

HIGHLIGHTS · 2026

- **5G Innovation Hackathon 2026:** two KIIT teams selected in the national Top 35 by the Department of Telecommunications, Govt. of India; prototypes funded, presented at IISc Bengaluru (Sept 2026) and showcased at India Mobile Congress 2026.
- **JSAP–Optica Joint Symposia 2026, Sapporo, Japan:** Saanvi presented in person and received the JSAP–Optica Travel Award; Saurav presented online (Sept 2026).

SELECTED PUBLICATIONS & PATENTS

S. Dass, A. Base, K. Agarwal, A. R. Nayak. An in situ stereo-imaging system for characterizing behavior, distributions and aggregations of meso- and macro-zooplankton in diverse aquatic environments. Limnology and Oceanography: Methods, 2026.

S. Patankar, S. Jena, P. Dash, **S. Dass**. Optimization of a D-shaped PCF using multiphysics for sensing applications. JSAP–Optica Joint Symposia, 2026.

S. Jena, S. Patankar, P. Dash, **S. Dass**. Surface plasmon activated D-shaped ZEONEX PCF as bio-compatible refractive index sensor. JSAP–Optica Joint Symposia, 2026.

R. Jha, **S. Dass**, J. N. Dash. Solid core photonic crystal fiber modal interferometer based micro-g accelerometer. Indian Patent 448718, 2023.

R. Jha, **S. Dass**. Optical fiber microphone/hydrophone based on cascaded single-mode fiber tapers. Indian Patent 567538, 2025.

Full list of 14 journal papers, book chapter and patents: sdlabkiit.github.io/#publications

Join SDlab

We are a young group looking for curious people who like building things that work: optics on the bench, code on the data.

- Ph.D. positions in holography, stereo imaging, fiber sensing
- Project positions and B.Tech. minor/major projects
- Skills we value: optics, Python/MATLAB, image processing, COMSOL, hardware tinkering

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Send a short CV and a paragraph on what you would like to work on.